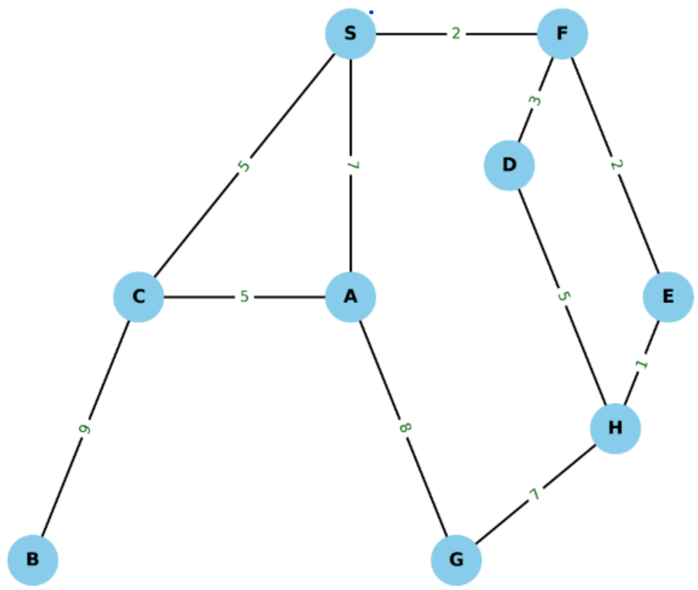
CSE422: Artificial intelligence

Assignment 01 [Spring 2026]

1. For the following graph, **S** is the **starting node**, and **G** is the **goal node.** Heuristic values are given in the table on the right.



Node	Heuristic	Node	Heuristic
S	12	D	8
A	14	E	15
B	16	F	9
C	10	H	17

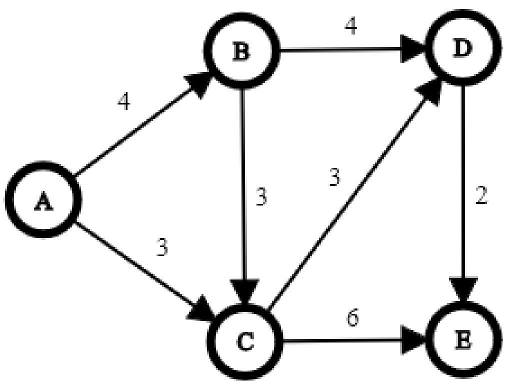
- Simulate Greedy Best-first search (GBFS) on the graph to find out <u>**the path to the goal**</u>, as well as <u>**the cost of the path.**</u>
- Simulate A-star search to find the optimal path to the goal, and compare it to GBFS output.

2. Imagine you have a **4×4** grid, where an agent's goal is to move from the **top-left corner** (0, 0) to the **bottom-right corner** (3, 3). The movement costs are: **horizontal or vertical = 1, diagonal = √2.** You are given TWO heuristics:

- $h_1(n) =$ **Manhattan distance** from node **n** to goal
- $h_2(n) =$ **Euclidean distance** from node **n** to goal

Which of these heuristics will expand fewer nodes in this grid? Show using proper simulations.

3. Design an **admissible** heuristic table for this graph. Consider **A** to be the source and **E** to be the goal.



4. Suppose you have three heuristic functions h1, h2 and h3. Among these h1 and h2 are admissible but h3 is inadmissible. You have decided to create several new heuristic functions defined as follows:

- h4(n) = 0
- h5(n) = 2 × h2(n)
- h6(n) = (h1(n)+h2(n))/2
- h7(n) = max(h1(n), h2(n))
- h8(n) = min(h1(n), h3(n))
- h9(n) = max(h2(n), h3(n))

Now answer the following questions and justify.

a. Which two heuristics are possibly inadmissible?
b. Among h6 and h7 which one is dominant?
c. In your opinion which heuristic is the best?

5. Suppose you are inside a grid-based maze, where each grid holds a treasure. The value of the treasure is given by an integer. When you are on a grid, you can collect the treasure of that grid, and you can also inspect the treasures in all neighbouring grids (i.e., the grids with which your current grid shares an **edge** or a **corner**). If you start from any of the **corner grids** (i.e. top-left, bottom-left, top-right, or bottom-right), can you find the highest valued treasure hidden in the maze using the **hill-climbing algorithm**? Explain why or why not. You can use the following grid given as an example.

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

6. State whether the following statements are TRUE or FALSE with a brief explanation. A short explanation is compulsory, just writing TRUE or FALSE will not yield any mark.

a. Simulated annealing's probability of denying worse solutions increases as the search progresses.
b. Simulated annealing guarantees better performance than hill climbing in all types of search spaces.
c. Simulated annealing becomes equivalent to hill climbing when the initial temperature is very high.
d. When optimizing a function with many local maxima of similar values, random restarts combined with hill climbing will never increase the likelihood of finding the global optimum.
e. A hill-climbing algorithm that stops after a fixed number of iterations will never find the global optimum if the number is large enough.

7. Select food items to pack in the lunchbox so that the total volume does not exceed 1 liters and the total nutritional value is maximized.
Lunchbox Volume Limit: **1 liters**
Food Items:

- Sandwich: Volume = 0.5 liters, Nutritional Value = 60
- Apple: Volume = 0.3 liters, Nutritional Value = 30
- Yogurt: Volume = 0.5 liters, Nutritional Value = 50
- Granola Bar: Volume = 0.2 liters, Nutritional Value = 40
- Carrot Sticks: Volume = 0.3 liters, Nutritional Value = 35

a. Encode the problem, so it can be solved using any **local search algorithm such as Hill Climbing.**
   [Hint: Encoding is similar to 0/1 knapsack genetic algorithm encoding, i.e. a string of binaries. However, in local search we need to have a START state. What will be the start state for this problem?]
b. Explain how you would generate neighboring solutions for this local search method.
   [Hint: For each state, a neighboring state can be found by adding one more food item. But what will happen when we are at the maximum volume limit?]
c. Using the given problem setup, demonstrate the main difference between Simulated Annealing and Hill Climb algorithm. Assume, Temperature parameter = 100 currently.